INTERIM FEDERAL SPECIFICATION

PLIERS: HOG RING STAPLE (UPHOLSTERER'S); BRAKE REPAIR (BRAKE SPRING): AND HOSE CLAMP (WIRE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers pliers used for installing upholstery fastener rings, automotive brake springs and self-locking wire hose clamps (see 6.1).

1.1.1 Federal specification coverage. This specification does not include all types, classes, styles, and sizes of pliers, but is intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, styles and sizes. Pliers shall be of the following types, classes, styles and sizes as specified (see 6.2):

Type I. Hog ring staple (upholsterer's).

Class 1. Straight jaws.
Class 2. Angled jaws.

Type II. Brake repair (brake spring).

Class 1. For heavy trucks and buses.

Style 1. With groove in end of handle.
Style 2. With groove in end of one handle, and socket or cup, spring remover and replacer on end of other handle.

Class 2. For passenger cars and light trucks.

Style 1. With groove in end of handle.
Style 2. With groove in end of one handle, and socket or cup, spring remover and replacer on end of other handle.

Type III. Hose clamp (wire).

Class 1. Swiveling jaw faces.
Class 2. Integral jaw faces.

Style 1. Without slip joint.
Style 2. With slip joint.

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2. APPLICABLE DOCUMENTS.

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of pliers which are otherwise in accordance with the requirements of this specification.

3.2 Material. Unless otherwise specified hereinafter, the pliers shall be steel, of such composition and quality that when properly heat-treated where necessary will produce tools conforming to the requirements of this specification.

3.3 Handles. Handles shall be shaped to afford a comfortable handgrip and shall be free from rough edges and sharp corners. The inner surfaces of the handle adjacent to the joint shall be rounded or beveled to prevent accidental pinching of the hands. The outer handgripping surfaces of the handles shall be smooth, knurled, or impressed, except that handles for type III hose clamps shall be knurled, impressed, or scored to minimized hand slippage.

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3.4 Joints. The pliers shall have riveted joints, except that type III, class 2 style 2, pliers shall have a bolt and nut. There shall be no excessive sidewise movement, play, or other indication of looseness of either half of the tool when the pliers are open or closed.

3.5 Jaws. The jaws shall operate through the maximum spread of the tool in a smooth and uniform manner without binding.

3.6 Finish.

3.6.1 Surface. All surfaces shall be substantially free from pits, nodules, burrs, cracks, and other detrimental defects. Surfaces not usually ground or buffed shall have a surface finish conforming to good manufacturing practices.

3.6.2 Coating. Unless otherwise specified (see 3.7 and 6.2), plier shall have one or a combination of the following coatings or plating at the option of the contractor (see 4.4.5) :

3.6.2.1 Oil-treated, natural. The coating shall be an oil-treated, natural finish free from excess oxide scale.

3.6.2.2 Oxide coating. The coating shall consist of a heavy chemically produced oxide.

3.6.2.3 Phosphate coating. The coating shall consist of a heavy chemically produced phosphate.

3.6.2.4 Bright nickel. The coating shall be electrodeposited nickel, and shall be not less than 0.00015 inch thick.

3.6.2.5 Chromium Plate. The finish shall be electrodeposited metals consisting of nickel, followed by chromium, the minimum thickness being 0.0002 inch for nickel and 0.00001 inch for chromium.

3.6.2.6 Zinc. The coating shall be electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.6.2.7 Lacquer. The coating shall be sprayed or dipped lacquer with corrosion-resistant properties.

3.7 Marking. The pliers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacturer may be readily determined, and with the plier size, if applicable. Pliers shall be marked before coating (see 3.6.2) so that the tools will have a complete unbroken coating.

3.8 Type I pliers, hog ring staple (upholsterer's). Type I pliers shall be forged from steel of a quality which when properly heat-treated shall comply with the applicable requirements. The rivet shall have a Rockwell C hardness reading of at least 30.(see 4.4.4). The nose shall have a Rockwell C hardness reading of at least 40 (see 4.4.4). The jaws shall have smooth grooves of a sufficient depth to firmly hold the rings. The jaws shall have a minimum opening of 1 inch.

3.8.1 Class 1, straight jaws. Type I, class 1 pliers shall have straight jaws in line with the longitudinal axis of the handle. The pliers shall be similar in design and conform to the dimension shown on figure 1.

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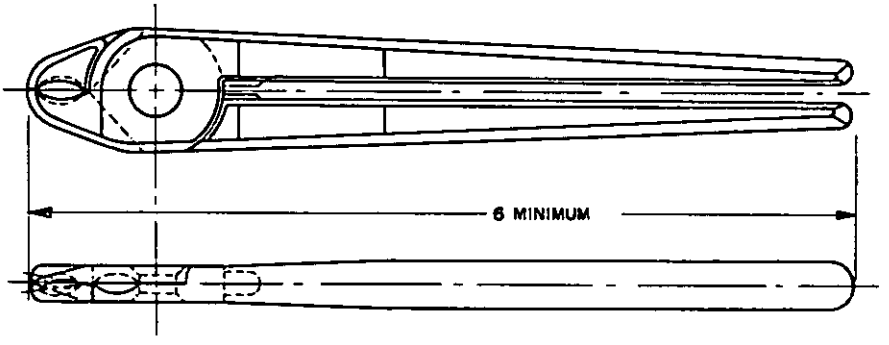


Figure 1. Type I, class 1 pliers, hog ring staple, straight jaw

3.8.2 Class 2, angled jaws. Type I, class 2 pliers shall have the jaws offset 60 degrees as measured from the centerline of the jaws and centerline of the handles. The pliers shall be similar in design and conform to the dimensions shown on figure 2.

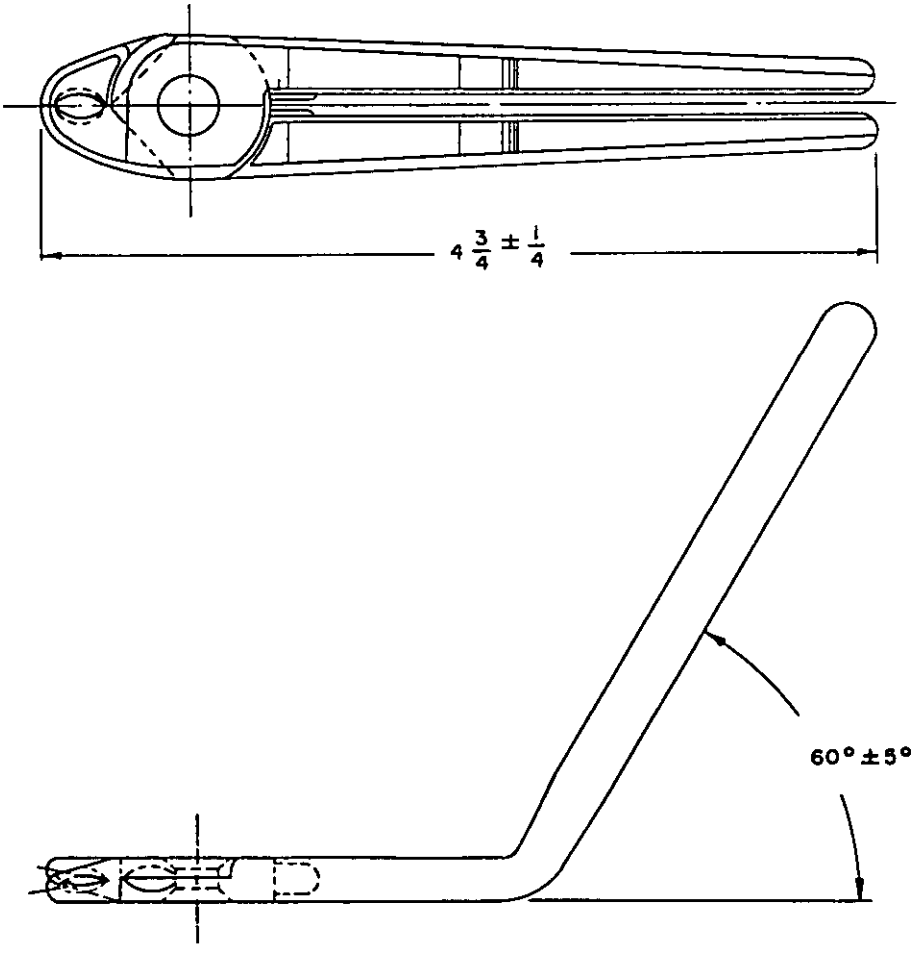


Figure 2. Type I, class 2 pliers, hog ring staple, angled jaws.

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3.9 Type II, brake repair (brake spring). Type II pliers shall be forged from steel with the working ends hardened to a Rockwell C reading of 48-52 (see 4.4.4). At the option of the manufacturer, the plier pivot shall be either a rivet or nut and bolt. The pliers shall be provided with a hook jaw or suitably slotted jaw, and a bent or curved jaw, suitable for removing and replacing brake springs. Type II pliers shall conform to the dimensions shown in table I.

Table I. Type II pliers, brake repair (brake spring)

Class	Overall length (Min.)
1	19 Inches
2	12 Inches

3.9.1 Class 1, for heavy trucks and buses. Type II, class 1 pliers shall be suitable for use on brake springs for heavy trucks and buses.

3.9.1.1 Style 1, with groove in end of handle. Type II, class 1, style 1, brake repair pliers shall be furnished with a groove or other suitable means in the end of one or both handles for quick replacement of the brake springs, and shall be similar to figure 3.

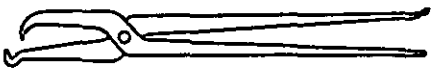


Figure 3. Type II, classes 1 and 2, style 1, pliers, brake repair (brake spring) for trucks, buses, and passenger cars.

3.9.1.2 Style 2, with groove in end of one handle, and socket or cup, spring and replacer on end of other handle. Type II, class 1, style 2, brake repair pliers shall be furnished with a groove or other suitable means in the end of one handle, and a socket or cup, on the end of the other handle, suitable for quick replacement of the brake springs, and shall be similar to figure 4.

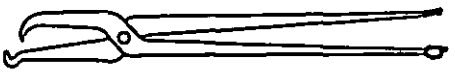


Figure 4. Type II, classes 1 and 2, style 2, pliers, brake repair (brake spring) for trucks, buses, and passenger cars.

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3.9.2 Class 2, for passenger cars and light trucks. Type II, class 2, pliers shall be suitable for use on brake springs for passenger cars and light trucks.

3.9.2.1 Style 1, with groove in end of handle. Type II, class 2, style 1, brake repair pliers shall be in accordance with 3.9.1.1.

3.9.2.2. Style 2, with groove in end of one handle, and socket or cup spring remover and replacer on end of other handle. Type II, class 2, style 2, brake repiar pliers shall be in accordance with 3.9.1.2.

3.10 Type III hose clamps (wire). Type III pliers shall be suitable for installing and removing hose clamps of the self-tightening spring wire type, used on automotive hose and industrial applications, up to and including 2 inch I.D. hose. The plier jaws shall open to p provide not less than a 2 inch clearance between the jaw faces (measured at the center of the faces). The center of each jaw face shall have a hole drilled or dimpled approximately 7/32-inch diameter by 3/32 inch (minimum) deep to preclude slipping of the wire clamp ends when being gripped. Two V-shaped notches extending from the outside edge of each jaw face to the drilled or dimpled hole shall be provided similar to that shown on figure 5 or 6. These notches shall be suitable depth or depths to assure a positive grip on each wire size of hose clamp within the capacity of the pliers.

3.10.1 Jaws and handles. The jaws and handles shall be of steel, fabricated by either forging or stamping. The jaw and handle may be of one piece or a jaw section and a handle section may be joined by riveting. When two-piece construction is used,the jaw section shall contain the plier pivot joint and shall have a shank extending not less than two inches beyond the center of this joint to provide for attachment of the handle by riveting. When riveted, at least two rivets spaced no less than 3/4 inch apart shall be used. The gripping surfaces of the handles shall be not less than 3/8 inch wide. Jaws and handles shall be of such quality steel that will withstand the load test specified in 4.5.3.1.

3.10.2 Class 1, swiveling jaw face. Type III, class 1 pliers shall be equipped with jaw faces that can be rotated in either direction not less than 180 degrees to permit convenient removal or installation of hose clamps in various positions. Each jaw face shall contain two V-shaped notches with one being shallower than the other. The shallow notch shall be best suited for gripping the smaller range of wire sizes while the deep notch shall be best for larger wire sizes. The faces shall be of casehardened steel to minimize wear. The jaw ends shall be designed to prevent accidental detachment of the faces but shall allow rotation of the faces by hand. The faces shall be spring loaded to minimize accidental rotation after being set to the desirex working position. The pliers shall be similar to figure 5 and shall conform to the overall length shown.

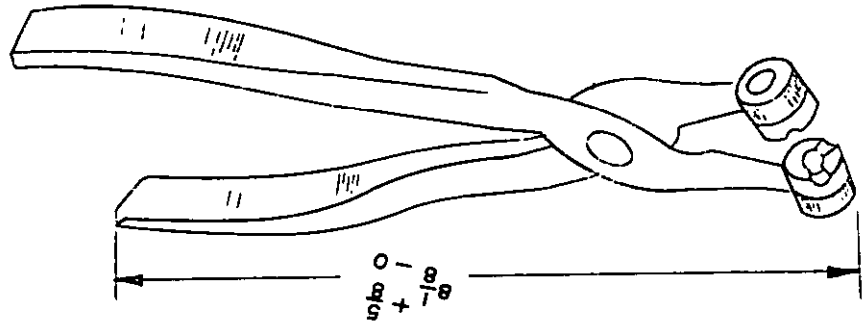


Figure 5. Type III, class 1 hose-clamp plier, swiveling jaw face.

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3.10.3 Class 2, integral jaw face. Type III, class 2 pliers shall have jaw faces that are integral with the jaw sections. The pliers shall conform to the overall length shown on figure 6 and shall be similar to figure 6 except that a slip joint is provided on style 2 pliers (see 3.10.3. (see 3.10.3.2).

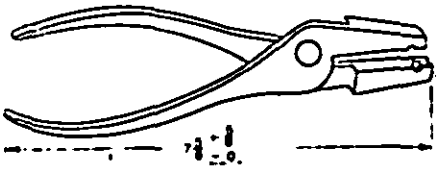


Figure 6. Type III, class 2 hose-clamp plier, integral jaw face.

3.10.3.1 Style 1 without slip joint. Type III, class 2, style 1, hose clamp pliers shall be furnished with a pivoted joint, not of the slip-joint type. To minimize wear, the jaw faces shall be either hardened to a Rockwell C reading of not less than 44 (see 4.4.4), or shall be casehardened steel. Stamped steel pliers shall be either chromium or nickel plated.

3.10.3.2 Style 2, with slip joint. Type III, class 2, style 2 hose clamp pliers shall be provided with a slip joint. The jaw faces shall be hardened to a Rockwell C reading of not less than 44 (see 4.4.4) to minimize wear.

3.11 Workmanship. The workmanship shall be first class in every respect. The pliers shall be free from rust, laps, cracks, blisters, and other defects which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

Table II Sampling Data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	One each	II	2.5	Per hundred units	4.3.1
Dimensional examination	One each	S-3	2.5	Per hundred units	4.3.2
Testing	One each	S-2	1.5	Per hundred units	4.4
Preparation for Delivery	One container	S-2	4.0	Per hundred units	4.5

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4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table III.

Table III Defects List

Design	Not as specified (3.8, 3.9 and 3.10)
Material	Not as specified (3.2)
Finish	Not as specified (3.6)
Coating	Not as specified (3.6.2)
Workmanship	Not as specified (3.11)
Marking	Not as specified (3.7)

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3, 4.4.3.1, 4.4.4 and 4.4.5 as applicable.

4.4.1 Type I pliers. The test rings used for compression tests shall be of various sizes and the type hog rings staples designed for use with type I pliers. A test ring of the largest size hog ring staple within the tool's capacity shall be properly inserted in the jaws of the pliers. A satisfactory compression and clinch for each of these ring staples shall be sufficient to determine the acceptability of the pliers. Upon completion of these compression tests, the pliers shall not show any evidence of breakage or permanent set or deformation.

4.4.2 Type II pliers. Pliers shall be tested for load in the following manner. A platform scale similar to figure 7 may be used for making load tests. A 1-inch steel spacer shall be held between the jaw faces while applying the load. The load test as specified in 4.4.3.1 shall then apply. Upon release of pressure and removal of the spacer, the pliers shall not show any evidence of cracks, breakage, or distortion, nor shall there be any evidence of deformation of the plier tips.

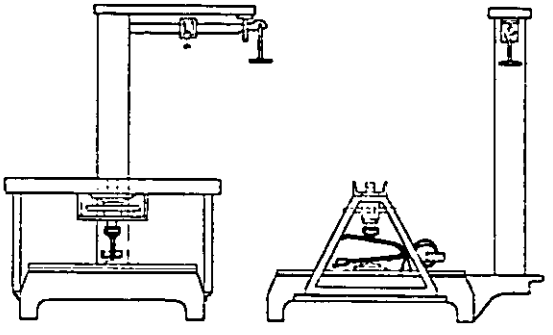


Figure 7. Apparatus for applying test load to handles of Type II pliers.

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4.4.3 Type III pliers. A self-locking type of wire-hose clamp having a 2-inch I.D. and 3/16-inch-wire diameter shall be used for the test ring during the contraction test on all type III pliers. This wire clamp shall be placed into the plier jaws and compressed manually until the jaw faces come within 3/8-inch of closing shut (measured at the outermost ends of the jaws). The pliers shall be held in this closed position for a period of 60 seconds, then the pressure shall be released and the wire clamp removed. If any slippage of the wire clamp ends occurs during the ring compression test period, of if the wire clamp should slip out of the plier's slots or grooves, this shall be cause for rejecting the sample pliers. After release, a visual examination shall be made to determine evidence of permanent set or deformation of the plier jaws or handles. Evidence of permanent set or deformation shall be cause for rejection. Hardness values shall be in accordance with 3.8, 3.9, 3.10.3.1 and 3.10.3.2.

4.4.3.1 Load test. The pliers under test shall be supported in a manner suitable for applying a load of 450-inch-pounds. The load shall be applied approximately 1-3/4-inches from the extreme end of the handles. A 1/4-inch (minimum) spacer shall be held between the jaw faces while applying the load. The actual load applied to the handle shall be calculated by dividing the 450-inch-pounds by the distance between the center of the rivet or bolt joint and the line of action of the load. If the permanent set is in excess of 1/16-inch, or if any rivet or bolt shows signs of weakness, or any part shows damage, the sample shall be rejected.

4.4.4 Hardness tests. Rockwell hardness tests shall be conducted in accordance with method 243 of Federal Test Method Standard No. 151.

4.4.5 Coating or plating tests. Pliers shall be tested for conformance to 3.6.2.1 through 3.6.2.7 as applicable.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table IV. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Table IV . Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per containers is more or less than required. Net weight exceeds requirements.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified (see 6.2) the hog ring, brake repair, and hose clamp pliers shall be preserved, packaged and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, table V is considered as part of and used in lieu of that portion of table I of MIL-H-15424 covering pliers, hog ring staple, brake repair and hose clamp.

Table V.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Hog ring staple, (Upholsterer's)						
Brake repair (Brake spring)						
Hose clamp (Wire)						
up through 6 inches	P-2, P-6	I	6		50	300
6 through 8 inches	P-2, P-6	I	6		48	288
8 inches and over	P-2, P-6	I	1		144	144

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I hog ring staple, (upholsterer's) pliers are intended for use in compressing fastener rings in upholstery.

6.1.2 Type II brake repair, (brake spring) pliers are intended for use in removing and replacing automotive brake springs.

6.1.3 Type III pliers, hose clamp are intended for use in removing and installing self-tightening wire hose clamps of the types used on automotive and industrial hoses.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type, class, and style of plier required (see 1.2).

(c) Size of plier required (see applicable tables).

(d) Coating, if particular type is required (see 3.6.2).

(e) Selection of applicable level of packaging and packing required (see 5.1).

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6.3 For Civil Agency Procurements. When level B packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

6.4 The following listing indicates new items or classification changes of pliers which are in Federal Specification GGG-P-480a:

GGG-P-480a	GGG-P-00474
Type IV - Hog ring staple (upholsterer's).	Type I - Hog ring staple (upholsterer's).
Class 1 - Straight jaws.	Class 1 - Straight jaws.
Class 2 - Angled jaws.	Class 2 - Angled jaws.
Type V - Brake repair (brake spring).	Type II - Brake repair (brake spring).
Class 1 - For heavy trucks and buses.	Class 1 - For heavy trucks and buses.
	Style 1 - with groove in end of handle
Style 2 - not previously covered	Style 2 - with groove in end of one handle, and socket or cup spring remover and replacer on end of other handle.
Class 2 - For passenger cars and light trucks.	Class 2 - For passenger cars and light trucks.
	Style 1 - with groove in end of handle.
Style 2 - not previously covered	Style 2 - with groove in end of one handle, and socket or cup, spring remover and replacer on end of other handle.
Type VI. - Hose clamp (wire).	Type III. - Hose clamp (wire).
Class 1. - Swiveling jaw faces. Class 2. - Integral jaw faces.	Class 1. - Swiveling jaw faces. Class 2. - Integral jaw faces.
Style 1. - Without slip joint. Style 2. - With slip joint.	Style 1. - Without slip joint. Style 2. - With slip joint.

GSA GC 67-18879

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